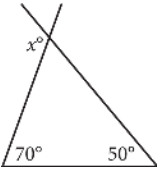


Question ID: 36200a38

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



In the figure above, two sides of a triangle are extended. What is the value of x ?

Answer

- A. 110
- B. 120
- C. 130
- D. 140

Correct Answer: B

Rationale

Choice B is correct. The sum of the interior angles of a triangle is 180°. The measures of the two interior angles of the given triangle are shown. Therefore, the measure of the third interior angle is $180^\circ - 70^\circ - 50^\circ = 60^\circ$. The angles of measures x° and 60° are supplementary, so their sum is 180° . Therefore, $x = 180 - 60 = 120$.

Choice A is incorrect and may be the result of misinterpreting x° as supplementary to 70° . Choice C is incorrect and may be the result of misinterpreting x° as supplementary to 50° . Choice D is incorrect and may be the result of a calculation error.